NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

L26.1 TianJi Close Reading and Hypothesis-Generation Mini-Trial

Group Rapid Trials Lab

Status Promoted

Period 2026-10-19 16:23 — 2026-10-24 21:24

Stage conclusion

Project report

L26.1 TianJi close reading and quick trial of hypothesis generation

ID: L26.1 · Status: Exploring · Candidate for promotion — awaiting PI review: the quick trial has been judged "worth using" per the criteria, with only the deployment handoff remaining (start approved by the PI on 10-19, relayed via L26 master control; launch plan archived)

Decision term definitions (defined before the start, 10-19)

- **Not considered at the time**: No entry with the same mechanism exists in the "list of hypotheses actually considered at the time" sealed by that track's lead (entries that differ only in wording while sharing the same mechanism do not count as new). Judged by that track's lead against the sealed list.
- **Worth testing**: That track's lead believes that, had it been thought of at the time, they would have been willing to spend at least one short test or one diagnostic to distinguish it.

- **Invalid hypothesis**: Meets any one of the following: the mechanism is physically incorrect; no test capable of distinguishing it can be provided; or it contradicts existing data on that track or established facts.
- **Blind evaluation**: The evaluator sees a list in which generated hypotheses and human hypotheses are mixed together, shuffled, and stripped of source labels; unblinding takes place only after evaluation is complete.
- The denominator is the total number of hypotheses submitted by the generator, with no post-hoc exclusions.

Close-reading key points

Source: full HTML text of the original preprint, arXiv:2603.27738v1. The local copy is kept in a non-published directory, because the arXiv non-exclusive license does not allow reposting the full text. Section numbers are cited below. 1. **Architecture** (§2): Based on AgentScope, divided into two parts. - Hypothesis generation: Researcher agents (configurable count, domain division of labor, and number of rounds) + Moderator + Chief Scientist. In the first round, paper titles and abstracts are retrieved before proposing hypotheses; in subsequent rounds they rebut each other; the Moderator scores each round on four dimensions: scientificity, plausibility, novelty, and validity; finally, the Chief Scientist selects a single hypothesis. - Hypothesis verification: Meta-Planner handles planning and state verification after each step, creating Workers on demand (wps_configurer, fnl_processor, wrf_real_executor, wrf_main_simulator, trajectory_analyzer, etc.) and invoking four categories of tools (WRF driver, tensor analysis, visualization, file/terminal/Python). Divided into simple and complex modes. 2. **Evaluation of hypothesis generation** (§3.1): Only 1 topic was tested (abnormal typhoon track), with 3 researcher agents over 6 rounds. The evaluation consists solely of scores assigned by the Moderator agent itself (e.g., Alice rising from 33 to 38, approaching full marks from Round 3 onward). - No human expert review, no comparison against direct generation by a single LLM, and no external metrics for coverage or novelty. - The main text does not list the verbatim text of the final hypothesis (possibly only in Figure 2, which could not be read from the HTML). - **Conclusion: Its hypothesis generation has not undergone independent evaluation.** 3. **Case 1: Squall line and soil moisture** (§3.2): 1 run each for two WRF setups, no ensemble. - The hypothesis under test is "dry soil strengthens the cold-pool gust front", yet the result is "cold-pool features significantly weakened, with core temperature deficit reduced by 20.4%", while the main text still claims to "strongly validate" it. - The passage on Figure 5 then shifts to claiming "the cold pool weakens, but gust-front lifting efficiency is enhanced" (-1.52 K → -1.21 K). - The supposed "autonomous discovery of a ~280 km southward shift of the main convection" is merely mechanistic speculation without a discriminating experiment. - Runtime was ~55 minutes with 164 API calls in total, automatically fixing 3 types of errors. One of them was "changing e_vert from 32 to 34 before re-running real.exe": a metgrid level mismatch should normally be fixed by changing num_metgrid_levels, whereas changing e_vert alters the model's vertical resolution, i.e., alters the experimental setup. **Questionable; the original paper did not provide the namelist, unverified.** 4. **Case 2: Typhoon In-Fa and +2 K SST** (§3.3): Likewise 1 run each. - Modification method: Added 2.0 K to SKINTEMP across all time steps in met_em, and set sst_update=0. - Result: Maximum southward track deviation of -0.3387° (~38 km), minimum sea-level pressure 944.2 → 924.7 hPa. Based on this, the agent "corrects the human linear northward-deflection hypothesis" and proposes

"intensity–track coupling". - Concern (a): The caption/description for Figure 10 states "warm anomaly beneath the warm core and cold anomaly around it" and also states "southward shift of ~280 km". Neither matches "uniform +2 K across the entire domain" or "maximum southward deviation of 0.34°". The 280 km figure is identical to that in the squall-line section, suggesting a copy-paste error. - Concern (b): SKINTEMP over land is also the land surface skin temperature, so adding 2 K across the entire domain simultaneously alters initial land-surface values. If met_em contains a separate SST field, the sea surface temperature used by real.exe might not have changed at all. The original text does not clarify whether only ocean grid points were modified; **unverified**. - A 0.34° track difference in a single deterministic simulation is not compared against the spread of a perturbed ensemble, making it impossible to distinguish "signal" from "chaotic noise". - Runtime was ~130 minutes with 180 calls in total; "vertical levels 38→34" similarly altered model settings to accommodate the data. 5. **Simple tasks** (§3.4): All 4 post-processing and plotting tasks succeeded (minimum pressure, track overlay, total precipitation RAINC+RAINNC, 10 m divergence). This is the most solid part, though low in difficulty. 6. **Public materials**: GitHub (created 2026-03-24, latest push 2026-07-23, 14 stars, **no license declared**). Contains only test outputs, README, and figures; no system source code, prompts, or namelists were found. A footnote in the paper says the code is planned for "subsequent open-source" release. 7. **Author-stated limitations** (§4): Only short-term mesoscale processes were tested; long-term climate and multi-sphere coupling remain unassessed. Our additions: n=1, no ensemble; no human review; definition of success is self-judged by the system; no human gatekeeping when altering model settings to fix errors.

Platform Lead's claim	Verification result
We are more complete in scale, human gatekeeping at key nodes, review/guards/tickets, and real supercomputer operations	**Holds**. TianJi only ran 2 cases, 1 pair of runs each, apparently on a single machine or small cluster; there is no review and no permission tiering; "zero human intervention" happens to include letting the agent modify e_vert and vertical levels on its own, which are operations that alter the experiment; our guards and mandatory review rules are designed specifically to block such operations.
It may be stronger at "systematically generating candidate hypotheses"	**Partially holds, needs qualification**. It does have a structured generation workflow (retrieval → multi-role hypothesis proposal → mutual rebuttal → scoring → chief selection), for which our platform lacks a corresponding fixed workflow. However, it ultimately **selects only one** hypothesis rather than delivering a candidate list with discriminating tests, and its generation quality has not been independently verified. What is worth borrowing is the workflow; its effectiveness is unproven.
It may be stronger at "fast single-track closed-loop iteration"	**Holds**. The two cases took ~55 minutes and ~130 minutes respectively from planning to report generation, fully automated from WPS to WRF to diagnostics. Every step on our side involves resource requests and registration, making a single round slower. The trade-off is that its speed is built on the absence of gatekeeping.

Platform Lead's claim	Verification result
It has public evaluation while we do not	**Needs correction**. What it made public consists only of self-scoring curves and self-reported accounts of 2 cases, not an independent evaluation, and the cases contain internal contradictions (see 3, 4). It is true that we lack a public system-level evaluation; however, it lacks a credible evaluation as well and cannot serve as a benchmark.

What was tried and results

Date	Action taken	Result
10-19	Close reading of original preprint full text, queried knowledge base, checked GitHub repository	See above; launch plan written, approved by the PI
10-19 16:27	Asked the WRF track to set the problem and seal its list of hypotheses from that time; execution protocol finalized	Sealed list submitted: Akron weak high-wind reproduction issue, 12 items, see wrf track for path, sha256 …81c0 verified matching, unopened
10-19	The original problem statement was not passed along with the result; asked the WRF track to resubmit it	Resubmitted, sha256 …b648 verified matching
10-19 16:27–16:38	Problem statement: Akron downburst WRF reproduction showing weaker high winds (R16.4, settled). Generated over three rounds following the TianJi method: G 19 mechanisms + D 7 diagnostics; single-model control group S 14 items	Sealed list (H 12 items) was opened only after generation completed; pre-opening hash recorded
10-19 16:45–17:05	Total of 45 items across G/S/H standardized in wording, shuffled, and submitted for blind evaluation; then unblinded	See next section

Quick trial results (10-19, numbers generated programmatically from the blind evaluation sheet)

Group	Count	New and worth testing	Invalid	Invalid share
G: TianJi-style deliberation (3 researchers × 3 rounds + moderator/chief)	19	5: G10 d03 underlying surface/spin-up, G12 boundary-layer erosion of stable layer, G13 DCAPE/sub-cloud moisture chain, G15 extreme air-parcel trajectory and pressure decomposition, G17 microphysical cooling	1 (G7 vertical levels, contradicts "bit-for-bit identical ZNW")	1/19 = 5%
S: Single-model one-shot enumeration (no retrieval)	14	3: S5 d03 static fields, S9 undisclosed author code modifications, S12 binary search via warm-start from authors' pre-storm snapshot	1 (S6, same as G7)	1/14 = 7%
H: WRF track's sealed list from that time (for calibration)	12	0 (expected 0, calibration passed)	0	0%

- **Per criteria**: Group G has "new and worth testing" 5 ≥ 1, and invalid 5% ≤ 50%, **judged worth using**.
- **Interpretation of decision terms**: When judging "not considered at the time", the evaluator checked against **both** the sealed list **and** the work logs from that time. The reason is that the work logs covered more possibilities than the sealed list (such as global attribute comparison, ZNW, station displacement, and phase lag), which the sealed list lumped broadly into items 4, 10, and 12. This is stricter than the definition written prior to start (checking against the sealed list only). Under the reading that checks only against the sealed list, there would be even more new entries, leaving the judgment unchanged.
- **Both readings side by side** (added 10-19, by script):
- "Not considered at the time" per the pre-start definition, checked against the sealed list H1–H12, judged by the lead using the mechanism correspondence table written prior to blind evaluation; the alternative reading is judged against the work logs.
- Under both readings, Group G has 5 "new and worth testing" items (G10, G12, G13, G15, G17) and Group S has 3 (S5, S9, S12); the results are identical.
- Reason: For entries where the two readings diverge (G2, G4, G7, G9, G16, etc., absent from the sealed list but present in the work logs), wrf judged all of them as "not worth testing", so the counts are unaffected.
- The judgment takes the pre-start definition (sealed list) as primary, and both readings judge it "worth using".

- **Coverage** (paraphrased from the evaluator's exact words): Both confirmed causes (WRF version, and 10 m wind diagnostic changing across versions) were covered in both Groups G and S.
- **Three qualifications verified by the lead** (the basis was written before the blind evaluation): 1. **Limited marginal gain**: The control group also passed the criteria; deliberation produced 2 more "new and worth testing" items than one-shot enumeration. In terms of cost, deliberation used 7 sub-agent calls totaling ~400,000 tokens and ~7 minutes, whereas the control group used 1 call totaling ~48,000 tokens, a factor of ~8.4. 2. **Deliberation downgraded a true cause**: The confirmed cause "10 m wind diagnostic changes across versions" was ranked last in G18, because the Round 2 order-of-magnitude estimate (U10/U(k1)≈0.8–0.9) conflated "the ratio of 10 m wind to lowest-model-level wind within the same version" with "the difference in 10 m diagnostic between two versions". The control group directly stated "10 m diagnostic modified across versions" in S8. Following the troubleshooting order given by the deliberation would have delayed finding it. 3. **Most discriminating new items singled out by the evaluator** were split evenly between the two groups: X21 = G15 (trajectory and pressure decomposition), X41 = S12 (warm-start binary search, missed by the deliberation group), X01/X02 = G10/S5 (d03 underlying surface).
- **Conclusion is strictly limited to** this problem (1 settled WRF reproduction issue), this workflow (sub-agents of the same model family playing multiple roles; retrieval limited to public literature), and this evaluator (that track's lead, who was also the decision-maker at the time). Do not extrapolate to "multi-agent deliberation is generally superior to direct prompting".

Deployment recommendations (submitted to L26 master control, which reports to the Platform Lead)

- **Format**: Package into a "candidate hypothesis list" task brief template or skill for leads to use when "reproduction does not match / diagnostics are stuck"; do not build into the production bot.
- **Workflow** (TianJi modified based on these results): 1. 2–3 perspectives each perform a one-shot enumeration, plus one single-pass enumeration without retrieval, taking their union as the draft 2. Conduct only one round of mutual rebuttal, **permitting only additions, merges, and flagging of doubts, while prohibiting deletion or downgrading of untested hypotheses based on estimates** (lesson from the downgrading error in this trial) 3. Must produce discriminating experiments, e.g., of the type "binary search via warm-start from authors' snapshot" 4. For each item, specify the mechanism, discriminating test, data, and cost 5. Submit to that track's lead to check off "already considered / worth testing / invalid" against the work logs.
- **Cost**: 0 SU, ~400,000 tokens, within 30 minutes.
- **Parts not adopted**: TianJi's "chief selects only one" and "moderator self-scoring". In this trial, scoring failed to serve as a filter and instead suppressed a true cause through faulty estimation.
- **By-product for the WRF track**: 8 "new and worth testing" items from the blind evaluation sheet (G10, G12, G13, G15, G17, S5, S9, S12). The WRF track lead has already seen them during evaluation; R16.4 will decide on its own whether to pursue them.

Open items and next steps

- The quick trial is complete. Remaining: L26 master control decides whether to report the "deployment recommendations" to the Platform Lead to open a ticket; the public citation list needs to add TianJi (arXiv:2603.27738) and Schumacher et al. 2023 MWR (doi:10.1175/mwr-d-22-0103.1). These two items are outside the write scope of this sub-project and are handed over to L26 master control to add.

Literature basis

See Section 4 of the launch plan. There is no literature card for TianJi itself in the knowledge base, only source notes and a close-reading card; for hypothesis-generation literature (AI Scientist, Co-Scientist, multi-agent debate), **no cards were found** in the knowledge base.

External basis for the fact-audit deployment recommendations (added 10-20): Kim, Geng, …, Neubig, Welleck 2026, *Into the Unknown: Benchmarking AI agents on open scientific research problems*, arXiv:2610.19159v1 (CC BY 4.0; code github.com/prometheus-eval/into-the-unknown, Apache-2.0). Literature card (created by litwiki on 10-20, with item-by-item PDF page numbers); close-reading card; main text read from PDF (HTML version failed conversion after Table 1), page numbers refer to physical PDF pages. - **What it supports**: 1. Errors by agents regarding "what has been completed and what the numbers are" are a widespread phenomenon, not unique to us. Across 135 submissions from 9 frontier agents, 114 failed completeness checks; the most common failures were discrepancies between text and code, contradictions between numbers and records, and claiming to have done something that was not done (§4, PDF p. 6). 2. The root behaviors are "lenient self-validation" (58 of 64 cases) and "unauthorized scope reduction" (21 cases) (§5.1, PDF p. 7). Our K1 ("learning task completed, acceptance passed") falls into this category. 3. **Writing necessary conditions first and then checking item by item against those conditions** reliably reproduces expert judgments on "whether requirements were met": 41 of 41 items covered, with complete agreement (Appendix A, Table 4, PDF pp. 15–16). This supports "having an independent check against pre-written conditions before a lead reports completion / acceptance passed". 4. Single-pass automated evaluation is not stable enough; multiple evaluations should be run and aggregated (Table 2 caption, PDF p. 6; Appendix B, PDF pp. 16–17). - **What it does not support**: 1. Different setting. In their setup, an agent independently writes an entire paper without human gatekeeping; in ours, the lead reports step by step with human and multi-model review gatekeeping. 114/135 cannot be taken as the error rate of our reports. 2. Their conditions are written by domain scientists (the original paper authors), whereas ours are written by the lead, without the same guarantee of strictness. 3. Their evaluators have access to complete code and run logs; our evidence is scattered across conversation databases, tickets, and cluster logs. Therefore, the v2 template must first specify the readable evidence scope and submission timestamp. 4. It does not judge scientific value (Appendix A.2: rubrics achieved 75% coverage and 64% agreement on "scientific quality and impact" comments); our audit gate likewise does not replace review. - Division of labor between the two papers: Agentic Scientist (§3.5) shows that "independent audit per round" can serve as part of a workflow, but does not quantify its effectiveness; this paper provides the types and frequencies of completeness failures, along with reliability data for "condition rubrics".

Stop category

Question answered (closure awaits the PI's decision)

Merged task (10-19, approved by the PI): Agentic Scientist close reading + "fact audit" quick trial

Fact audit quick trial · header (defined before the start, 10-19 17:04, unchanged thereafter)

- source: The "fact audit" stage of The Agentic Scientist (Sakana AI, arXiv:2610.18933); commissioned by L26 master control
- question: Can an auditor without prior knowledge of the answer key check the reports submitted by the lead to the PI item by item and catch known false statements without excessive false positives?
- criteria: Known-error recall ≥ 1/2 **and** false positives ≤ 1/3 of all flags → worth using; otherwise abandon. Also record the time spent per report.
- time_limit: ≤1 day
- resource_limit: 0 SU, 0 Colab; read-only, no modifications to any reports or files
- status: Worth using
- conclusion: Worth using (pre-registered primary reading, both criteria just passing the threshold): recall 2/4 = 50%, false positives 4/17 = 24%; when "unverifiable" flags are counted as false positives, the rate is 11/17 = 65%, judged abandon. It also caught 4 true errors outside the answer key.

Decision term definitions (defined before the start)

- **Known errors**: Entries in the answer key, including false statements in the 3 designated samples as well as errors I independently verified myself after sampling and before the auditor returned results. The answer key was finalized before the auditor returned results, with its sha256 recorded.
- **Recall**: Number of known errors explicitly flagged by the auditor as "incorrect, unsubstantiated, or conclusion exceeding evidence" whose referent matches an answer-key entry (same sentence or same fact) ÷ total number of known errors.
- **Flag**: Each "problematic" judgment issued by the auditor, counted per item; duplicate flags referring to the same fact count as 1 item.
- **False positive**: A flag for which, upon my verification, the flagged content was supported by evidence at the time the report was sent and was not actually wrong. "Unverifiable" flags are counted separately

and count as neither false positives nor hits. Errors newly discovered by the auditor and verified by me to be genuinely wrong are counted as true positives and listed separately in the statistics.
- **Evaluation method**: Calculate once under the primary reading; another reasonable reading counts "unverifiable" as false positives, and both are reported together in the conclusion.

Sampling method (defined before the start)

- **Population**: Replies signed by the Platform Lead between 10-15 00:00 and 10-19 16:00 across the Platform Lead's two root conversations (P20 master control, and opening new projects and sub-projects) with body text ≥ 800 characters.
- **Sampling**: Exclude the 3 designated samples along with messages within ±2 positions adjacent to them (to avoid sampling corrections of the same matter), then use Python `random.seed(20261019)` to randomly draw 7 reports from the remaining population without cherry-picking.
- **Format delivered to the auditor**: 10 original texts, unannotated, in shuffled order, with only the sent timestamp attached.
- **Confidentiality**: Original texts and the answer key are stored on the local machine only; this page and the statistics file record only sample IDs, dates, and statistics.

Agentic Scientist close-reading key points

Original paper: Lange, Grillotti, Yamada, Charakorn, Ha (Sakana AI) 2026, *The Agentic Scientist: Orchestrating Coding Agents for Automated Scientific (Co-)Discovery*, arXiv:2610.18933v1, CC BY 4.0. Read from the full arXiv HTML text, local copy. The HTML version has no page numbers; section numbers are cited throughout below, and PDF page numbers were not checked.

1. **Four-stage workflow** (§3): Task specification → Parallel exploration → Retroactive evidence generation → Grounded writing, each stage running as an independent process. The orchestrator enforces stage-gate "rubric" thresholds.
2. **Task contract** (§3.1): A task comprises an initial program (deterministic baseline), a search evaluation program, an optional final evaluation program, and a machine-readable contract. Drafting passes through several gates: - An independent reviewer agent first reviews the experimental design - The harness runs a smoke test - An "alignment judge" checks scientific fidelity and confirms that search evaluation and final evaluation are mutually independent and **use disjoint data** - Only versions that pass all gates may proceed to the next stage.
3. **Contracted research loop** (§3.3): - The ideation agent is **read-only** and must submit a structured idea package: title, falsifiable hypothesis, ordered implementation plan, and target rubrics; packages missing any field are rejected outright. - The execution agent may only **append** candidate versions, working within a fixed evaluation budget following a "modify–test–verify" loop. - Two gates operate independently: "all rubrics passed" is required for contract completion, whereas updating the best program requires correctness and a **strictly higher** score.
4. **Baselines and ablations** (§3.4): - Ablations are proposed based on the "code diff between the selected program and the initial program", with each ablation isolating a single innovation - Filtered by another

agent for relevance, feasibility, and **single modification** - Can be run across multiple seeds - Programs are frozen during post-hoc analysis; any proposal that would retrain or modify a program is rejected by programmatic checks.

5. **Fact audit** (§3.5): - The writing agent simultaneously produces a "claim → source" mapping - Each revision round is checked by an **independent auditor agent** on a read-only copy of the workspace, flagging numbers, comparisons, or conclusions that exceed the evidence or contradict raw data - Additional deterministic checks cover structure, source links, compilation, and length.
6. **Audit effectiveness is not quantified**: The paper nowhere reports recall or false-positive rates for the auditor agent (confirmed by the lead searching the full text for "audit"). The paper itself provides two counterexamples: - §4.3: During human review, "ED-PPO significantly outperforms BP-PPO" still had to be revised to "on par", unsupported citations had to be removed, and omitted primary references had to be added. Table 2 shows that prior to submission, humans inserted 17.5% and deleted 10.7% of the text. - §6.3, Figure 16: In 2 out of 3 runs of Pi, the agent unilaterally altered the test-time evaluation protocol (substituting majority voting for pass@1). Neither the evaluation program nor the rubric gate caught this; it was discovered only through **post-hoc human audit**. - The paper cites Luo et al. 2026: LLM auditors discovering issues from logs and code are far more reliable than inspecting the paper alone (§A Related Work).
7. **Evaluation methodology and limitations**: - §5: Across 70 research runs, o4-mini AI review scores are used as a proxy for paper quality, which the authors explicitly state is "not evidence of scientific correctness"; only 1 paper (ALIFE 2026) underwent external peer review, and it was submitted only after major human revision. - §8: Validated only in computational machine learning, which does not demonstrate transferability to domains requiring physical experiments; the completeness of verification depends on the evaluation program and rubrics, and constraints not written into the rubrics cannot be caught.

Dimension	Agentic Scientist	TianJi	Our platform	What we have extra / what we lack
Stage gates	Rubric gate at every stage, uniformly enforced by the orchestrator (§3, §3.3)	Self-checked by Meta-Planner after each step (§3.2), no external gates	Launch plan gate, five-category mandatory review, gates requiring the PI's sign-off; automated initial check only verifies deliverable completeness	**Extra**: Human and multi-model blind review gating real decisions. **Lacking**: Machine-checkable stage gates; most gates rely on natural language and the lead's diligence
Task contract	Executable contract: evaluation program, disjoint data, smoke test (§3.1)	Natural-language instructions only	Task brief, --accept checklist, pre-registered criteria, decision term definitions, experiment card	**Extra**: Operational definitions of decision terms, mandatory annotation for post-hoc criteria changes. **Lacking**: Executable evaluation programs; initial check does not verify content

Dimension	Agentic Scientist	TianJi	Our platform	What we have extra / what we lack
Constrained execution	Container sandbox, read-only ideation, append-only candidates (§3.3)	Agent can modify namelists (e_vert, vertical levels) on its own without gatekeeping (§3.2–3.3)	Guards (SSH, single submission per root, no recursive deletion), write scope, read-only rules, resource application and job registration	**Extra**: Hard safety for real supercomputer operations. **Lacking**: Per-loop workspace isolation and "append-only" version discipline
Baselines / ablations	Automatically proposes single-change ablations, multi-seed (§3.4)	Only 1 pair of deterministic runs per case, no ensemble (§3.2–3.3)	Experiment card requires specifying the "single modification"; ensembles and controls are manually designed by each track	**Extra**: Domain-specific physical control design. **Lacking**: Automatic generation of ablations from code diffs and programmatic checking of single modifications
Fact audit	Claim → source mapping + independent audit every round (§3.5), but effectiveness unquantified and major human edits still needed (§4.3, §6.3)	None; conclusions contain internal contradictions (§3.2, §3.3)	Only review (five categories) and automated initial check; **routine reports submitted by leads to the PI have no routine fact audit**	**Lacking**: Claim → source verification for routine reports. This track's "fact audit" quick trial tests whether this step is worth adding

Fact audit quick trial results (10-19; numbers generated by script)

- **Process**:
- Criteria and sampling method finalized at 17:04
- 10 samples: 3 designated by the commissioning party + 7 randomly drawn (seed 20261019, out of 104 candidates)
- Answer key frozen at 17:09, 4 items, of which K4 was identified by the lead
- Auditor was an unbriefed independent sub-agent, started at 17:06, prohibited from reading conversation logs and subsequent discussions
- Upon return, adjudicated item by item by the lead.
- **Results**: 17 flags = 6 true positives + 4 false positives + 7 unverifiable.
- Recall 2/4 (K1 "learning completed, acceptance passed" and K4 "dispatched" were hit; K2 misidentified figure and K3 boundary sample were missed).
- False positives 4/17 = 24%.
- **Judgment**: Under the primary reading, **worth using**, with both criteria just passing the threshold. Alternative Reading B counts "unverifiable" as false positives, yielding a false-positive rate of 65% and a judgment to abandon. Both readings are presented side by side.

- **Its strengths**: Caught 4 true errors outside the answer key that I had also overlooked myself, for example:
- One report attributed a stream interruption to a gateway restart, whereas ticket records showed the root cause was an earlier out-of-memory event
- "Three uploads" was actually four batches
- Two numbers were overstated, or an aggregate scope was misapplied to a single item.
- **Its weaknesses**:
- Without background context, it cannot catch "misidentified object" errors (K2).
- 3 false positives stemmed from the same timestamp-definition issue: the message header timestamp marks when the round started, not when it was submitted.
- 7 "unverifiable" flags occurred because the evidence resided in the conversation database, which the auditor was prohibited from reading.
- **Post-hoc note (17:35, discovered during adjudication, criteria unchanged)**: The message header timestamp is the start time of the current round. K4 is counted solely for "dispatched"; "ticket opened" fell within the current round and does not count as an error.
- **Limitations**: Only 10 samples and 1 auditor (from the same model family as the lead), and only the Platform Lead's reports were audited. Do not extrapolate to "auditing can catch the majority of errors".

Fact audit deployment recommendations (submitted to L26 master control, which reports to the Platform Lead)

- Serve as an **optional gate**, used only for "completed / acceptance passed / deployed live / ready to include in report" conclusions submitted by a lead to the PI. An unbriefed auditor checks numbers, paths, ticket IDs, and statuses item by item against primary evidence.
- **What the task brief must provide**:
- The **submission timestamp** of the current round, not just the start timestamp
- Permitted evidence scope: tickets, files, job records; whether to open the conversation database is to be decided by the platform.
- **Three categories returned by the auditor**: "incorrect / exceeds evidence / unverifiable". Adjudicated by the lead; reports are not modified automatically.
- **Cost**: 0 SU, ~10–20 minutes for 10 reports.
- **What it does not do**: Does not replace mandatory review, and does not use Gemini (this is judgment work).
- **10-20 revision**: Revised into the v2 template based on Into the Unknown (arXiv:2610.19159), with a comparison table. There are three categories of changes: (1) write necessary conditions first; (2) subdivide error categories, splitting "unverifiable" into "evidence unreadable" and "evidence not found"; (3) anchor on the submission timestamp, open read access to the current conversation's history prior to the submission timestamp, and perform a second audit on decisive flags.

Lead check (10-19 17:50, master control)

- Read the unblinding results, the fact-audit statistics, the answer key, and the adjudications item by item. Both trials are judged "worth using" per criteria. Fact audit recall passes the threshold only under some readings (1/3 fails when counting only the 3 samples designated by the commissioning party), so the conclusion is marked "barely passing". Hypothesis generation deployment is changed to a single-model checklist template without multi-role deliberation.

Fact audit v2 live test (10-21, approved by the PI under item 10: first live-test v2 as an optional gate and submit it for review)

Header (defined before the start, 10-21 14:45, unchanged thereafter)

- source: This track's L26.1 Fact Audit v1 (10-19) and v2 template; approved by the PI on 10-21 under item 10 of the L26 overall report
- question: When conducting a round of fact auditing using the v2 template, can it still catch known errors with few false positives? Are the two flaws targeted by v2 (timestamp-definition false positives, and "unverifiable" items due to inability to read conversations) reduced?
- criteria: Criteria carried over from v1: known-error recall ≥ 1/2 **and** false positives ≤ 1/3 of all flags → v2 may continue to be used as an optional gate; otherwise abandon v2 and revert to the v1 conclusion.
- time_limit: ≤1 day
- resource_limit: 0 SU, 0 Colab; read-only
- status: Worth using
- conclusion: Retain as a non-mandatory optional gate for continued trial use (wording after review): the two pre-registered readings pass the criteria (recall 4/8, exactly at the threshold; definite false positives 1/11), but the strictest v1 reading and the "counting only the 3 commissioning-party samples" reading both fail; results come from the same batch of samples and lean optimiztic; "unreadable evidence" did not improve.

Design

- **Same samples as v1**: The same batch of 10 reports R01–R10, with the auditor replaced by a new, unbriefed independent sub-agent.
- This was done to allow a direct comparison with v1: identical reports, changing only the template.
- Trade-off: The samples are not new, so this cannot speak to v2's performance on other reports.
- **Answer key v2**:
- v1 answer key K1–K4
- 4 true errors newly found by the v1 auditor and verified by the lead: R04 quota increase, R05 stream-interruption attribution, R07 NIST coefficient of variation, R08 upload count
- 8 items in total, frozen prior to the run, with sha256 recorded.
- **v2 modifications executed per template**:
- Provide the submission timestamp (from the task record) for the relevant round of each report
- Provide the auditor with the messages in each report's conversation prior to the submission timestamp (most recent 15 messages), exported by the lead
- Provide a list of necessary conditions: a single general checklist uniformly applicable to all 10 reports, rather than written individually per report, to avoid the lead baking the answers into the conditions
- Decisive flags (those that would alter "pass/fail") are double-checked once by a second auditor.

Decision term definitions (carried over from v1, with two additions)

- **Recall**: Number of known errors flagged by the auditor as an error category or exceeding evidence whose referent matches an answer-key entry ÷ 8.
- **False positive**: A flag for which, upon verification by the lead, the flagged content was supported by evidence at the submission timestamp and does not count as wrong. Both subcategories of "unverifiable" are counted separately and do not count as false positives.
- **Primary reading**: As above. **Alternative reading**: Count "unverifiable · evidence not found" as false positives ("evidence unreadable" still excluded).
- **Additional observations, not part of criteria**:
- Number of "unverifiable" items, compared against 7 in v1
- False positives caused by timestamp definition, compared against 3 in v1
- Agreement rate between the second audit and the first audit
- Whether K2 (missed in v1) is caught (now that prior conversation context is open for reading).

Fact audit v2 live test results (10-21, under review)

- Primary reading: Recall 4/8 = 50%, false positives 1/48 = 2%; per criteria, v2 **may continue to be used as an optional gate** (pending review verdict). Both the alternative reading and the two post-hoc supplementary readings also pass the criteria, though recall remains stuck near 1/2.
- Hit K2: The "misidentified object" missed in v1 was caught this time using the prior conversation context.
- Missed K1, K3, K5, K7.

- Newly caught 6 true errors outside the answer key.
- Second-review agreement rate: 9/11.
- 37 "unverifiable" items, mostly still due to inability to read the conversation database or commissioning records.

- Submitted for review by 3 reviewers.

- **Review verdict reached** (10-21 16:20). Seven revisions made per the comments: definite false-positive rate, added the readings that fail the threshold, noted the optimiztic bias, stated explicitly that "unverifiable" did not improve, unified v1/v2 accounting (new true errors 6→5), disclosed adjudication reasoning and hashes, and added "unauthorized scope reduction" to strict template evaluation. Rewrote header conclusion accordingly. Not recommended for promotion to a mandatory gate.

Citation snowballing expansion (10-22, PI directive to re-run snowballing across all sub-projects)

- Run and screening log on file.
- Card creation delegated to litwiki (no changes to conclusions or criteria before cards are returned).
- Preliminary impact: Judgment unchanged; if new cards contain "multi-agent vs. single-model" ablations, compare on the project page.
- Card check (10-22): Robin (Nature 2026, "A multi-agent system for automating scientific discovery") only performed ablations swapping individual roles to o4-mini and candidate comparisons against Deep Research, **without** a causally isolated "multi-agent vs. single-model" experiment under identical tools and budget; hypothesis quality was evaluated via LLM pairwise verdict + expert preference calibration, with candidate activity tested separately in vitro (PDF pp. 7, 14). The Herron 2026 hypothesis-generation review likewise only listed individual cases where debate/critique helped, without compiling controlled comparisons that could support a conclusion of "generally superior to a single model" (PDF pp. 11–13, 27). → Our trial's finding that "deliberation adds only 2 items at ~8.4 times the cost" serves as single-problem evidence addressing this gap; the conclusion remains strictly limited to this problem and workflow without extrapolation; judgment unchanged.

Key figures